The area of a polygon

Any polygon at all: cut it into triangles, add the pieces, done.

LESSON 50 | 1 × 40 MIN | GEOMETRY 8, TEMA 51 | STAGE 9 · 7.2

LESSON CLOCK

5' | 12' | 12' | 7' | 4'

Cut a pentagon into triangles	5 min
Decomposition	12 min
Compound shapes	12 min
Regular polygons	7 min
Homework	4 min

LEARNING OBJECTIVES

- Decompose a polygon into triangles and rectangles.
- Find the area of a compound shape by adding or subtracting.
- Find the area of a regular polygon from its apothem and perimeter.
- Find areas from coordinates on squared paper.

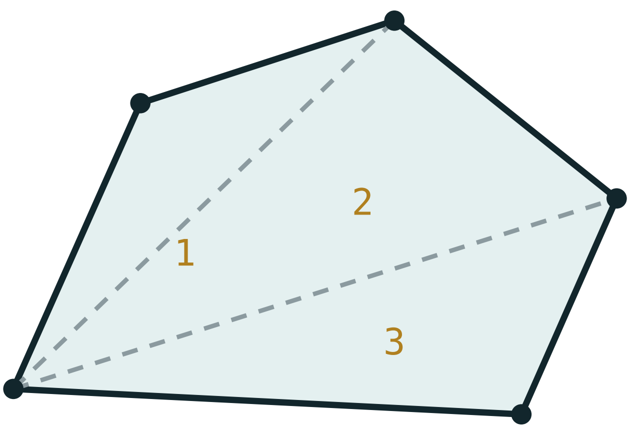
TEXTBOOK

National	Тема 51, pp. 119–121
Cambridge	Learner’s Book pp. 149–155
Workbook	Workbook 7.2

01 Explanation

Cut it into pieces you know

Any polygon can be cut into triangles by drawing diagonals from one vertex. By the additivity of area, the whole is the sum of the parts:



split into triangles, then add the areas

Diagonals from one vertex cut a polygon into triangles; add their areas to get the whole.

An n-gon splits into $n - 2$ triangles this way — the same count that gave the angle sum $180°(n - 2)$ in Quarter I. The two facts have the same source.

TWO STRATEGIES

Adding: cut the shape into rectangles and triangles and total them.
Subtracting: draw the smallest rectangle that contains the shape, then take away the corner pieces.
On squared paper the second is usually faster.

Regular polygons

A regular n-gon splits into n congruent triangles meeting at the centre. Each has base a (a side) and height r (the **apothem** — the distance from the centre to a side):

$$S = n \cdot \frac{1}{2} a r = \frac{1}{2} P r$$

where $P = na$ is the perimeter. A regular hexagon of side 6 has apothem $3\sqrt{3} \approx 5.2$, so $S = \frac{1}{2} \cdot 36 \cdot 5.2 \approx 93.5$.

APOTHEM, NOT RADIUS

The apothem reaches the **middle of a side**; the radius reaches a **vertex**. They are different lengths, and only the apothem belongs in this formula.

02 Worked examples

EXAMPLE 1

An L-shape is a 10×6 rectangle with a 4×3 rectangle removed from one corner. Find its area and perimeter.

1. $10 \cdot 6 = 60$

   The whole rectangle.

2. $4 \cdot 3 = 12$

   The piece removed.

3. $60 - 12 = 48$

   The area.

4. $P = 2(10 + 6) = 32$

   The perimeter is unchanged — the cut adds two edges and removes two of the same total length.

ANSWER

Area 48, perimeter 32

EXAMPLE 2 **Find the area of the polygon with vertices $(0;0)$, $(6;0)$, $(6;4)$, $(3;7)$, $(0;4)$.**

1. Split into a rectangle and a triangle.
   The horizontal line $y = 4$ does it.

2. *rectangle* $6 \times 4 = 24$

3. *triangle: base 6, height 3*
   From $y = 4$ up to $y = 7$.

4. $\frac{1}{2} \cdot 6 \cdot 3 = 9$

5. $24 + 9 = 33$

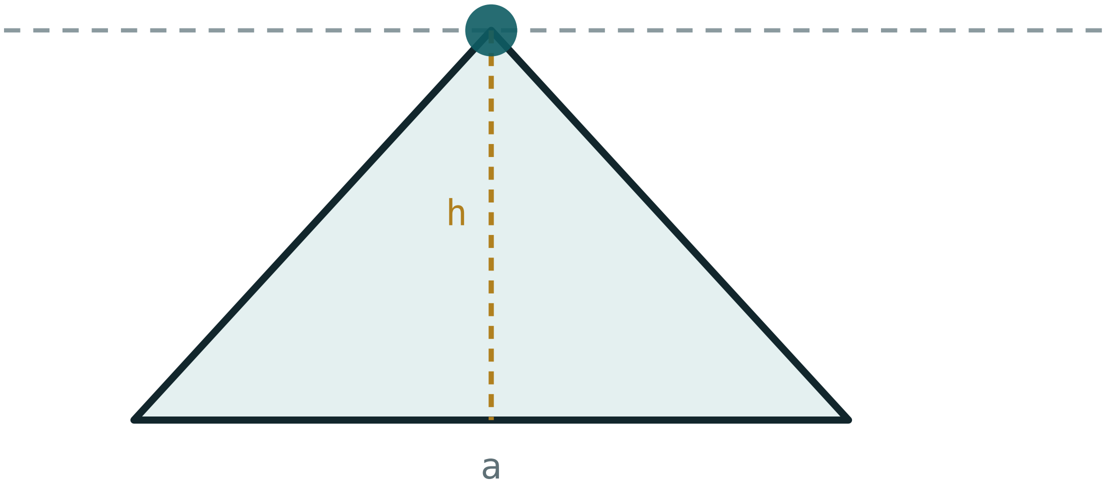
ANSWER

33

03 Interactive model

Change the base and height of one piece and watch how the total responds.

One piece of a decomposition



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Polygon	Ko‘pburchak	Многоугольник
Decomposition	Bo‘laklarga ajratish	Разбиение
Compound shape	Murakkab figura	Составная фигура
Regular polygon	Muntazam ko‘pburchak	Правильный многоугольник
Apothem	Apofema	Апофема
Enclosing rectangle	O‘rab turgan to‘rtburchak	Описанный прямоугольник
Subtraction method	Ayirish usuli	Метод вычитания
Cross-section	Kesim	Сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A pentagon splits into how many triangles from one vertex?

- 3
- 4
- 5
- 2

An L-shape is a 8×5 rectangle less a 3×2 corner. Its area is:

- 40
- 34
- 46
- 6

For a regular polygon, S equals:

- $P \cdot r$
- $\frac{1}{2} P r$
- $P + r$
- $n \cdot r$

The apothem of a regular polygon reaches:

- a vertex
- the middle of a side
- the opposite vertex
- the circumcircle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *An L-shape is a 6×4 rectangle less a 2×2 corner. Find its area.*

   ANSWER *20*

2. *A hexagon splits into how many triangles from one vertex?*

   ANSWER *4*

3. *Find the area of a shape made of a 5×3 rectangle and a triangle of base 5, height 2.*

   ANSWER *20*

4. *A regular polygon has perimeter 24 and apothem 4. Find its area.*

   ANSWER *48*

5. *An L-shape is a 9×5 rectangle less a 4×2 corner. Find its area.*

   ANSWER *37*

6. *Find the area of a triangle with vertices (0;0), (4;0), (0;3).*

   ANSWER *6*

7. *A regular hexagon has perimeter 36 and apothem 5.2. Find its area.*

   ANSWER ≈ 93.6

Medium · the standard question

7 PROBLEMS

1. *Find the area of the polygon (0;0), (8;0), (8;3), (4;6), (0;3).*

   ANSWER *36*

2. *A rectangle 12 × 8 has a 4 × 4 square cut from two opposite corners. Find the area left.*

   ANSWER *64*

3. *A T-shape is a 10 × 2 bar on top of a 4 × 6 column. Find its area.*

   ANSWER *44*

4. *A regular octagon has perimeter 40 and apothem 6. Find its area.*

   ANSWER *120*

5. *Find the area of the quadrilateral (0;0), (5;0), (7;4), (2;4).*

   ANSWER *20*

6. *A cross is made of five 3 × 3 squares. Find its area and perimeter.*

   ANSWER Area *45*, perimeter *36*

7. *A trapezium sits on a rectangle: rectangle 8 × 3, trapezium with parallel sides 8 and 4, height 3. Find the total area.*

   ANSWER *24 + 18 = 42*

Hard · stretch and reason

7 PROBLEMS

1. *Find the area of the pentagon (0;0), (6;0), (8;4), (4;8), (0;4).*

   ANSWER *44*

2. *A regular hexagon has side 8. Find its apothem and area.*

   ANSWER $4\sqrt{3} \approx 6.93$, area ≈ 166.3

3. *A room 6 m by 5 m has a 2 m by 1.5 m alcove added. Find the floor area and the cost of tiling at 120 000 so‘m/m².*

   ANSWER *33 m², 3 960 000 so‘m*

4. *A rectangle 10 × 6 has a triangle of base 10 and height 6 removed. Find the area left, and explain the answer.*

   ANSWER $60 - 30 = 30$ — the triangle is exactly half the rectangle

5. *Find the area of the quadrilateral (1;1), (7;2), (6;6), (2;5).*

   ANSWER *20*

6. *A regular polygon has 12 sides of length 5 and apothem 9.3. Find its area.*

   ANSWER ≈ 279

7. *Explain why cutting a polygon into triangles always gives the same total area, whichever cuts you use.*

   ANSWER Additivity says the parts of any one decomposition sum to the whole, and the whole does not depend on how it was cut — so every valid decomposition gives the same total.

07 Homework

Homework — 5 problems

Geometry 8, Тема 51, pp. 119–121. Draw the cutting lines before calculating.

1. *An L-shape is a 12 × 7 rectangle less a 5 × 3 corner. Find its area.*
2. *Find the area of the polygon (0;0), (7;0), (7;4), (3;7), (0;4).*
3. *A regular polygon has perimeter 30 and apothem 5. Find its area.*

4. *A T-shape is a 12×3 bar on a 4×5 column. Find its area.*

5. *A regular hexagon has side 10. Find its apothem and area.*